The Hillingdon Hospitals
NHS Foundation Trust

Maternal Antenatal Screening Tests

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Guideline Number	403
Version Number	5.1
Date Issued	19th December 2025
Next Version Date	8th January 2028
Approval Group	Maternity Document Review Group (Chair)
Approval Date	18th December 2025
Ratified Group	Clinical Outcomes and Effectiveness Group (Chair)
Ratified Date	18th December 2025

Change History

Version	Author	Summary of changes
4.0	Gillian Pearce, Maternity Incentive Scheme Lead Midwife	Full review including CO monitoring
4.1	Gillian Pearce	Changes in Appendix 4, processes for serology screening out of hours
5.0	Lucinda Hartstone	Full review
5.1	Gillian Pearce Helen Hosking	Updated information in 3.13 – regarding unbooked service-users and urgent screening tests Appendix 1 update for B12 levels and removal of TORCH screen terminology

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Contents

1	Aim/Purpose
2	Introduction
3	The Guidance
3.1	Designated Lead for Antenatal Screening in the maternity service
3.2	Offering Antenatal Screening tests
3.3	Pre-booking information
3.4	Booking
3.5	Haemoglobinopathies
3.6	Infectious Diseases
3.7	Down's Syndrome
3.8	Documentation of Antenatal Screening Tests
3.9	Process for requesting blood tests
3.10	Women who decline screening
3.11	Ultrasound scanning - Dating scan - Fetal Anomaly Scan
3.12	Screening test undertaken within appropriate time frame
3.13	System for ensuring that appropriate tests are undertaken when women book late or for an unbooked woman in labour
3.14	Known HIV and Hepatitis B at booking
3.15	Process for the review and reporting all results to women
3.16	Process for reporting results to other relevant healthcare professionals
3.17	Process for ensuring that women with screen positive test results are referred and managed within appropriate timescales
3.18	Process if a woman declines or does not attend an appointment
4	Associated Guidelines, Policies and Documents
5	Training Need Identified
6	Implementation and Monitoring

Appendices

Appendix 1	Timetable for Screening Tests
Appendix 2	Referral to Antenatal Screening Specialist for Positive Results or for Women who Decline Blood Test
Appendix 3	Process for samples outside of hours
Appendix 4	Flowchart for Managing Antenatal Screening Tests
Appendix 5	Low Papp-a letter

1. Aim/Purpose

The guideline explains the referral pathways to the relevant health professionals for pregnant women who screen positive for Human Immunodeficiency Virus (HIV), Hepatitis B, Haemoglobinopathies, Syphilis, and fetal abnormalities.

This guideline describes the process for ensuring all recommended antenatal screening tests are offered to all pregnant women. The aim of screening is to ensure that the appropriate care/interventions are offered to improve the outcome for mothers and babies. It is supported by individual guidelines stated in the section on associated guidelines, policies and documents.

The guideline will enable midwives to provide relevant information to pregnant women at any stage of their pregnancy and for those caring for the unbooked woman in labour. The guideline gives a brief overview of the process for engaging with relevant health professionals if a pregnant woman screens positive for HIV, Hepatitis B, Syphilis, Haemoglobinopathies or fetal abnormalities.

2. Introduction

The work of the UK National Screening Committee (UK NSC 2024) reviews screening evidence, making screening recommendations, advising ministers and the NHS in the 4 UK countries about all aspects of screening and supporting implementation of screening programmes. The Committee defines screening as a process for identifying apparently healthy people who may be at increased risk of a disease or condition. They can then be offered information, further tests and appropriate treatment to reduce their risk and/or any complications arising from the disease or condition.

All pregnant women regardless of gestation should be offered antenatal screening as recommended by NHS England screening (2023) and the National Institute of Health and Clinical Excellence (NICE 2021). The guideline aims to ensure that all recommended tests are offered, undertaken, reported on and actioned within an appropriate timescale.

The NHS England screening recommends that all women are offered screening for infectious diseases (HIV, Hepatitis B, Syphilis), haemoglobinopathies, Down's. Edwards and Patau's syndromes and an anomaly scan at 18-20 weeks of pregnancy. In addition NICE (2008) recommends screening for asymptomatic bacteriuria, anaemia, blood group and red cell alloantibodies.

Women identified in early pregnancy with an infection (HIV, Hepatitis B or syphilis) can be offered early treatment and appropriate care/interventions to provide optimal management of maternal health and reduce the risk of mother-to-baby transmission.

Identifying women at increased risk of having a baby with Down, Edwards or Patau syndromes or couples at risk of having a baby with a serious haemoglobin disorder gives the woman the option of having a diagnostic test.

Where a fetal abnormality is diagnosed, and the pregnancy is continuing the woman will be offered appropriate support and information. Women who opt for a termination of pregnancy will be offered appropriate care and support.

Identifying women who have been exposed to CO either because they smoke, a member of the household smokes, or the woman has been exposed to a faulty appliance or car exhaust. (Detailed in another guideline)

The guideline seeks to clarify the care pathway for offering, undertaking, communicating and acting on results. It also includes in Appendix 1 other antenatal tests taken due to clinical signs and symptoms and the referral pathway.

3. The Guidance

3.1 Designated Lead for Antenatal Screening in the Maternity Service

The designated Consultant Obstetrician lead and the Antenatal Clinic and Community Midwives Manager are responsible for ensuring that any screening recommendations from the PHE screening are implemented.

The maternity service has identified lead roles for all antenatal screening tests. These roles comprise assessment and care planning with the multi-professional team.

Screening test	Identified lead
Screening for Down's Syndrome	Antenatal and Newborn Screening Coordinator
Hepatitis B / C HIV Syphilis	Specialist midwife – Infectious diseases and sexual health
Sickle Cell, Thalassaemia	Specialist midwife – haemoglobinopathies
Anaemia, Blood group & RhD status Red cell antibodies	Sister Antenatal Clinic Team leaders of each community team
Blood glucose	Specialist midwife – diabetes
High chance results	Fetal Medicine specialist midwives
Ultrasound scanning	Senior Sonographer for maternity

The screening test leads are responsible for:

- Ensuring that antenatal screening guidelines reflect the NHS England standards
- Monitoring and/or audit screening outcomes as recommended by the UK NSC
- Data collection and reporting to the UK NSC, NHS England
- Attend the Antenatal Screening Group meeting

3.2 Offering Antenatal Screening Tests

All pregnant women will be given the opportunity at booking to be screened for specific conditions, which includes all tests recommended by the UK NSC Guidance.

UK NSC NHS England Guidance on screening conditions	The Hillingdon Hospitals NHS Foundation Trust recommended additional screening tests at booking
Antenatal screening midwives	Antenatal clinic or fetal medicine midwives
• Sickle Cell and Thalassaemia • Syphilis • Hepatitis B • HIV • Down's Edwards and Patau's Syndrome • Ultrasound scanning – including dating and anomaly scans • *Hepatitis C (not currently part of national screening programme but part of NWL sector-wide practice)	• Blood group & Rhesus D status & red cell antibodies • Timed random blood sugar or random blood sugar • Anaemia • Asymptomatic bacteriuria • CO monitoring • **Fetal medicine team as required (**See separate fetal medicine guideline)

3.3 Pre-Booking Information

Women should be directed to the STFYAYB QR code. If there are additional needs, such as adjustments for protected characteristics under the following characteristics are protected characteristics—age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex; in line with the Equality Act 2010, Section 4. Ethnicity and language barriers must not hinder or impede informed consent in the screening process; these will be addressed by contacting supporting agencies such as Language Line/translator on Wheels as necessary or accessing the Trust's public website.

Family members and friends should not be used for translating purposes.

3.4 Booking

The **booking midwife** must ensure that all women and birthing people can access the STFYAYB QR code and understand the screening tests offered before consenting to testing. They should also discuss the antenatal screening tests available for routine antenatal care. This will enable women to make an informed choice about having the tests or not.

3.5 Haemoglobinopathies, Sickle Cell and Thalassaemia

The blood test is offered preferably before 10 weeks gestation (UK NSC 2024) to enable prenatal screening. The midwife must request haemoglobinopathy testing on Cerner at the booking appointment of all women regardless of any known status (unless they decline). The family origin question must be completed as well. This is a request in Orders as part of the Booking Bloods Bundle.

If the mother is identified as a carrier based on her screening result, it will indicate whether the partner needs to be tested. Partner testing will be offered to identify couples with genotypes specified as requiring an offer to assess the risk to the fetus of inheriting a serious haemoglobinopathy. The midwife/doctor checking the blood results will inform the specialist midwife for haemoglobinopathies if a partner needs to be tested. This will be followed up by an email to the SCT specialist midwife.

3.6 Infectious Diseases

HIV, Syphilis, and Hepatitis B and C are infectious diseases, and the fetus is at risk of transmission.

3.7 Down's Syndrome, Edwards Syndrome and Patau Syndrome

The maternity service encourages referrals to the antenatal clinic team by **9+0 weeks gestation**. All patients should be given the QR code and the link to the National Screening Committee's publication '*Screening Tests for you and your Baby*' at the first pregnancy consultation. This then allows time for the nuchal scan appointment to be arranged, giving patients the option of the combined screening test for Down's syndrome, Edward's and Patau's (this test is performed between 11+2–14+1). All women referred by 9 weeks will be booked for a dating/nuchal scan before 14 weeks gestation. Women who are referred between 9 and 14 weeks of gestation may be offered a scan appropriate for the combined test if appointment capacity allows. The Combined Screening Test also tests for Papp-A. Later referrals will have the option of the Quadruple test (performed between 14+2 – 20+0 weeks). A screening blood test accompanies the first scan to ensure screening is completed if required.

Midwives need to counsel about Trisomy 21, 18 and 13. Women and birthing people have the option to accept testing for all conditions or to decline the test.
Any declines need to be documented in the maternity Cerner record.
The options are:-

- to choose not to have screening or
- to choose screening for Trisomy 21 (T21) and Trisomy 18 (T18) / Trisomy 13 (T13)
- to choose to screen for T21 only
- to choose screening for T18 / T13 only.

The combined test uses maternal age, nuchal translucency measurement (NT), and two biochemical markers to provide a risk assessment for the pregnancy. The maternal blood specimen should be taken at the time of the ultrasound scan. The first trimester combined test allows parents to make earlier decisions. The Combined Screening Test gives each twin an individual chance of T21/T18/13. A Quadruple test is offered if the patient is scanned after 14+1. Both tests can be offered for twin pregnancies, but the Quadruple test will only give one overall result for Down Syndrome for both twins. Quad screening in twin pregnancies poses a difficulty because one fetus may be affected and the other may not.

The Down's syndrome test is gestation-specific and can only be performed between 11+2 to 20 weeks. Women who present after 20 weeks of pregnancy will be too advanced in their pregnancy to be offered screening for Down's syndrome. An anomaly scan should be offered and, if accepted, booked as soon as possible.

Consanguinity Consanguineous relationships occur when a couple are genetically related and can be associated with adverse genetic and perinatal outcomes for affected offspring. Please ensure a lead sonographer fetal medicine anomaly scan is arranged for these pregnancies. Request on Cerner as FMU scan and give details.

3.8 Documentation of antenatal screening tests

The booking midwife must document antenatal informed consent on the electronic booking record on Cerner.

- This includes a discussion of routine blood tests: iron levels, timed random blood sugar, Syphilis, HIV, Blood group and antibodies, Hepatitis B and C, Thlassaemia/Sickle cell, and CO monitoring.
- This includes discussing Down's syndrome, Edward's, and Patau's, explaining the risk and at what gestation each test can be done. *It must be explained and discussed that this is a screening test and that it is not diagnostic; therefore, there is always a chance of the baby having chromosomal abnormalities at birth.*
- This includes a discussion on anomaly scans, explaining that the purpose of the scan is to identify structural abnormalities, the possibility that further tests may be offered, and limited views on women with a high body mass index (BMI).

3.9 Process for Requesting Blood Tests

- Complete blood and nuchal requests on Cerner for all consented blood tests, ensuring that the family origin question is completed.
- For all Down's syndrome screening please complete the request form for Down's blood test on Cerner at the booking appointment.
- All samples must be correctly labelled
- All routine bloods are sent to the pathology department
- All Down's, Edwards' and Patau's screening blood tests are done in the antenatal clinic and sent daily by courier service to Birmingham Women and Children's Hospital laboratory (BWC).

3.10 Women Who Decline Screening

If a woman declines screening for infectious diseases or other blood tests at booking, she should be counselled and reoffered screening again at 20 weeks. The opportunity to be screened later in the pregnancy should always be an option. The midwife or obstetrician can discuss screening tests at an antenatal appointment. If a blood sample has been received in the laboratory, e.g. for Hepatitis B and C. The woman decides later to be screened for any other infectious disease, e.g. HIV; a second blood test is not always necessary. The laboratory should be telephoned and asked to test the stored blood sample.

Women have the right to an informed choice, including the choice not to have screening tests. It is the midwife's responsibility to explore the woman's concerns about screening and to ensure that the woman understands why screening is recommended and the advantages and disadvantages of screening for her and the baby's health if she should screen positive. The booking midwife can refer women to the specialist midwife for counselling. The booking midwife needs to inform the specialist midwife of the reasons given by the woman who has declined the test.

3.11 Ultrasound Scanning

Dating scan

This scan is usually offered around 8-14 weeks gestation. The scan is performed with the aim of confirming:

- Pregnancy
- Gestational age
- Number of fetus

Fetal anomaly scan

This scan is offered between 18 – 20+6 weeks of pregnancy. Women booking after 20+6 can be offered the scan however they must be made aware that the accuracy of the test is

affected by advanced gestation. This scan is performed with the aim of detecting the following 11 major conditions in the fetus:

- Anencephaly
- Open Spina Bifida
- Cleft Lip
- Diaphragmatic Hernia
- Gastroschisis
- Exomphalos
- Serious Cardiac Abnormalities
- Bilateral Renal Agenesis
- Lethal Skeletal Dysplasia
- Edwards Syndrome (Trisomy 18)
- Patau's Syndrome (Trisomy 13)

The woman is referred by the sonographer who has undertaken the ultrasound scan to the Fetal Medicine Specialist Midwife for counselling. If the patient needs a Tertiary Centre Referral, this must be made on the day of the scan, and it can be done by the Fetal Medicine Specialist Midwife after discussion with a relevant obstetric consultant if required.

If the fetal anomaly scan cannot be completed at the time of the anomaly scan, an appointment to rescan within 2 weeks is offered, or sooner if there are any concerns.

3.12 Screening Test Undertaken within Appropriate Time Frame

See the table for screening blood tests that the midwife will routinely offer, the timetable, the action to be taken and who to refer to in the case of a positive result.

3.13 System for Ensuring that Appropriate Tests are undertaken when Women Book late or for an Unbooked Woman in Labour

Women who book after 12+6 weeks gestation must be offered all routine antenatal blood tests at the booking appointment.

Women who book at more than 14+1 but less than 20+5 weeks will have a discussion with the booking midwife (leaflet) on quadruple testing for Down's syndrome.

Unbooked women who present from 34 weeks gestation or in labour (at any gestation) will have a complete obstetric, medical, and social history taken by the midwife, which will include all routine blood documented on Cerner. It is essential that all screening bloods be taken with consent and that this is communicated on the SBAR when there is a shift change or transfer of care.

Women presenting in labour must be offered screening for HIV, Hepatitis B and C and Syphilis, Full Blood Count, Blood Group and Antibody Screening. Any screening test taken with consent after 34 weeks gestation must be marked urgent. Results will be available within 24 hours to the maternity services once the laboratory has received the sample. The process for HIV sample testing enables the result to be returned within four hours. For unbooked patients, take two bottles for serology testing out of hours as HIV and Syphilis and Hepatitis B and C can go to different centres for testing.

If the mother has booked elsewhere, it is important to retake the booking blood tests but also check with the booking hospital for screening results to avoid any delay in acting on positive results.

In cases where consent for screening is withheld during labour, it must be documented in electronic maternity health records, and screening must be offered again postnatally.

3.14 Known HIV and Hepatitis B and C at Booking

Where a prior diagnosis of HIV and Hepatitis B is documented, this must be recorded in the hospital notes and arrangements made for a prompt clinical evaluation and referral to the relevant specialist (Gastroenterology et cetera). The woman must be offered screening tests as normal.

3.15 Process for Reviewing and Reporting All Results to Women (Appendix 5)

Within 7 days of booking blood, results must be checked and actioned, ensuring they are updated on the patient's record. Go to Obs and Assessment, select the antenatal tab, and then select Review of Booking Investigations. Check all results present, correct, and complete, then ensure the results are endorsed.

On the main antenatal summary page, you will see the blood group. At the 16-week appointment, the midwife should check that the results have been actioned. Go to Obs and Assessment, antenatal, review of booking investigations, then on the grey date bar, change the date selection to a year ago - current, and you should see the entered results. Please note if you have selected an encounter prior to the encounter used for inputting results you will not see any results

All urine and blood test results are checked by the midwife/doctor before being documented in the maternity electronic health record. All normal results are reported to the woman and documented as communicated to the woman in the maternity electronic health record at the next antenatal visit—see Appendix 1.

Booking blood and urine results that are normal are recorded at the 16 week appointment.

- It is the responsibility of the midwife/doctor who obtains consent for the test to follow up the result or delegate responsibility to another midwife/doctor within 7 days of the blood or urine sample being taken and document as done.
- The booking midwife will contact the woman with any results (by letter or telephone call, depending on circumstances) that require urgent treatment or referral, such as low haemoglobin, antibodies, variant haemoglobins, raised blood sugar levels, or a Midstream urine sample indicating a urinary tract infection or group B Strep, *within 7 days of the tests being taken and documented as done.*
- The results checker will make an appropriate management plan for abnormal results. The midwife/doctor must discuss outcomes with the appropriate specialist midwife/ consultant obstetrician. The midwife/doctor must document the actions taken and the results on the electronic Message Centre on Cerner in Power Chart and enter them onto Cerner.
- Screen positive results for HIV, Syphilis, Hepatitis B and C; the laboratory will contact the specialist midwife, who will contact the woman.
- All Down’s blood test results are returned to the unit by courier except high-risk results, which are emailed. The specialist midwife will send a letter to the woman for all low-risk results. For high-risk results, the woman will be phoned and invited to attend an appointment for further discussion.
- The midwife checks the booking results and records screening results, especially positive results, on the Cerner Power chart maternity computer system. The results are to be checked again at the next antenatal visit, usually 16/40.

It is the duty of all healthcare professionals to ensure that the results are known and recorded and that they are discussed with the woman.

3.16 Process for Reporting Results to Other Relevant Healthcare Professionals

- The specialist midwife will arrange appointments with the relevant consultant obstetrician, paediatrician, or medical leads where appropriate.
- All referrals and actions will be recorded on the Maternal Health Record on Cerner.
- The ultrasonographer will report any fetal abnormalities detected on the anomaly scan to the Specialist Midwife for Fetal Medicine. If an obstetric review is required, the midwife specialist will arrange an appointment.

The woman is advised that if they have a rash illness or encounter a rash illness, they should inform their midwife or GP urgently and avoid contact with other pregnant women.

If the woman reports a generalised rash, then the laboratory must be asked to check the IgM as well as IgG for rubella to verify whether she has rubella or not.

3.17 Process for Ensuring that Women with Screen Positive Test Results are Referred and Managed within Appropriate Timescales

The laboratory will contact the specialist midwife with any positive infectious disease results. Any positive result will be actioned within 7 days of receipt. The failsafe position is that the person who took the blood will review the results and implement relevant action in conjunction with the specialist midwife/consultant obstetrician within 7 days.

All screen positive results are reviewed and actioned by the specialist midwife who will offer an appointment to see the specialist midwife to discuss the result and make the appropriate referrals to the multi-disciplinary team within 10 working days of the result being available. This is documented in the maternity health record.

All positive results are entered into Cerner with the woman’s consent, and an alert message with the diagnosis and need for infant follow-up is put on Cerner.

3.18 Process if a Woman Declines or Does not attend an Appointment

If a woman is referred and declines to go to see a specialist such as Genitourinary Medicine (GUM) then the obstetrician will liaise with the GUM consultant for a follow up management plan to be done in the Maternity Unit.

If the woman does not attend her appointment with the specialist (GUM, Gastroenterology), then the specialist midwife will liaise with the woman to find out the reasons and arrange a follow-up appointment.

The specialist midwife refers the Hepatitis B-positive woman (recently diagnosed or known) to the gastroenterologist within 6 weeks for an assessment and management plan. If the woman declines, the obstetric consultant/specialist midwife will liaise with the gastroenterologist for a follow-up management plan to be done in the Maternity Unit.

4. Associated Guidelines, Policies and Documents

- Haemoglobinopathies screening guideline and flowchart
- Sickle Cell disease and pregnancy
- Protocol and flowchart for a baby’s ‘at risk’ of a Major Haemoglobinopathy who decline Prenatal Diagnosis

- Antibodies during pregnancy
- Rubella screening in pregnancy
- Down's syndrome screening and diagnostic testing
- Referral when a fetal abnormality is detected
- Management and flowchart of Hepatitis B in pregnancy
- Management of Hepatitis C in pregnancy
- HIV infection in pregnancy
- Management of positive Syphilis Serology in pregnancy
- Diabetes pathway for abnormal blood glucose
- Gestational Diabetes diagnosis and antenatal care
- Pre-existing diabetes for type 1 and type 2
- FGR guidelines

5. Training Need Identified

Training for this guideline is provided as described in the Training Needs Analysis.

6. Implementation and Monitoring

All guidelines, protocols, and procedures are available on the intranet. All staff members are responsible for checking the Trust's intranet and ensuring that they refer to the most recent issue of the document. The weekly general information bulletin includes notification of the publication of new/updated documents.

The antenatal screening tests for infectious diseases, antenatal screening and haemoglobinopathies are reported as key performance indicators to NHS England quarterly. This clinical guideline, protocol or procedure will be monitored using local and National guidance and via the Trust Incident and Risk Management Policies. If risks/issues are identified, further specific monitoring will be considered, for example, clinical audit or risk assessment

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Appendices

Appendix 1: Timetable for Screening Tests
Appendix 2: Referral to Antenatal Screening Specialist for Positive Results or for Women who Decline Blood Test
Appendix 3: Hillingdon Antenatal Infectious Diseases Screening Pathway
Appendix 4: Flowchart for Managing Antenatal Screening Tests
Appendix 5: Low Papp-a letter

Appendix 1 - Timetable for Screening Tests

All results must be documented in the electronic maternal health record. The woman will be informed of the normal results at the next visit.

Test	When test Offered	Review results within	Action if abnormal	Refer to
Asymptomatic Bacteriuria (Red top urine) if dipstix indicates nitrites	Booking	3-5 days	The woman is treated with antibiotics/antifungals	GP by letter/ Obstetrician (in patient)
Anaemia FBC (Haemoglobin (Hb) < 10.5 at booking or MCV < 80) (lilac top) Ferritin (gold top) Folate (gold top) B12 (gold top)	Booking 28 weeks and 34 weeks	1 day	Iron therapy Vitamin C 500mg if Hb < 10.5 g/dl If Hb < 10 further studies, check for ferritin, folate and B12 The midwife should review diet and advise accordingly **Ferritin** < 50mcg/L in early pregnancy requires iron supplements, < 12mcg/L indicate iron deficiency, **Folate** **B12** normal range 71-165pmol/L **Platelets** < 150 repeat FBC with blood film **Guideline for the investigation and management of thrombocytopenia in pregnancy**	Refer to consultant obstetrician if < 10g/dl at booking Anyone who is microcytic (low MCH) must have a B12 and ferritin at next appointment to ensure patient is on correct supplements Abnormal results Refer to consultant. obstetrician Haematology clinic And refer to anaesthetist if < 100
Vitamin D (gold top)	Booking for diabetics, consider in high risk groups obstetric decision	3 days	25-50 Less than 25mcg	Pregnancy approved supplement/Vitamin D 10mcg Refer to obstetrician
Fetal DNA (Large pink top)	16 weeks	5 days	Cell free fetal DNA (cffDNA) in maternal blood for Rh negative result patients	
Blood group (Large pink top)	Booking & 28 weeks	1 day	If the mother is rhesus negative a letter and leaflet is sent to her. Discussed next visit	28 week appointment-Anti D given

Test	When test Offered	Review results within	Action if abnormal	Refer to
Red cell alloantibodies (pink top)	Booking & 28 weeks	7 days	Repeat bloods taken as indicated by laboratory	Refer to Lead Obstetrician and specialist midwife
Cytomegalovirus (CMV) IgG & IgM (rust top) Toxoplasmosis IgG & IgM (rust top) (previously known as TORCH screen)	Rash during pregnancy	If required days	Ask about immunisation history – ask the woman to confirm with GP surgery if she has had 2 doses of MMR Parvovirus IgG & IgM requested if signs of fetal anaemia present **See Rubella guideline**	Midwife Complete GP letter to give 2nd MMR
Hepatitis B/C (rust bottle top)	Booking	3-7 days	If positive refer to specialist midwife for infectious diseases / consultant Document on Cerner with alert for vaccine for baby **See Management and flowchart of Hepatitis B in pregnancy guideline**	Refer to obstetric medicine
Syphilis (rust bottle top)	Booking	3-7 days	Refer to specialist midwife Document on Cerner **See Management of positive Syphilis Serology in pregnancy guideline** If blood result is equivocal in labour	Refer to Consultant obstetrician and Cons paediatrician for management plan
HIV (rust bottle top)	Booking	3 days	Refer to specialist midwife/ consultant Document on Cerner **See HIV in pregnancy guideline**	Specialist midwife/ Lead Consultant Obstetrician
Sickle cell/ thalassaemia (lilac bottle top)	Booking Preferably before 10 weeks	3 days	If partner testing required (stated on lab report) – offer testing Seen by specialist midwife for haemoglobinopathies Offer prenatal diagnosis if appropriate See haemoglobinopathies screening guideline and flowchart	Specialist midwife/ Consultant obstetrician obstetric medicine

Test	When test Offered	Review results within	Action if abnormal	Refer to
Down's Syndrome, Edwards and Patau's Combined Blood test (rust top)	11+2 to 14+1	Abnormal results emailed to specialist midwife 10 working days check results	Counselling by specialist midwife. Offer prenatal diagnosis and further testing if there is a deviation in blood results PAPP-A less than 0.415 MoM **See Down's Syndrome, Trisomy 13 and 18 – Screening and Diagnostic Testing**	Specialist Midwife in genetics / Consultant obstetrician Discuss with obstetrician about growth scans
Down's Syndrome Quadruple test (rust top)	14+2 to 20+0	Abnormal results emailed to specialist midwife 7 days check results	Counselling by Fetal Medicine Specialist Midwife Offer prenatal diagnosis Abnormal AFP above 2.5 MoM – refer to for FMU sonographer led scan for spina bifida **See Down's Syndrome, Trisomy 13 and 18 – Screening and Diagnostic Testing**	by Fetal Medicine Specialist Midwife/ Consultant obstetrician high risk Antenatal screening scans
Glucose (grey top)	Booking GTT 28 weeks or earlier	1 day	*TRBS – Ate within 2 hours of test > 7.0mmol/l, Ate over 2hours of test > 5.5mmol/l* *Random blood sugar >7.0mmol/l* *GTT pre ≥ 6.0mmol/l* *Post ≥ 7.8mmol/l* **See pathway for abnormal blood glucose results**	Refer for GTT ASAP Refer to Diabetic specialist midwife
Anomaly scan	18-20+6 weeks	At time of scan	If anomaly scan is incomplete, a rescan in 2 weeks. Any abnormalities detected by Fetal Medicine Specialist Midwife **See Obstetric Ultrasound guidelines**	Consultant obstetrician/ tertiary unit as indicated, high risk AN screening scans

Blood tests taken as need arises, due to clinical situation				
Test	**When test Offered**	**Review results within**	**Action if abnormal**	**Refer to**
U&E's (gold top)	PET, Obstetric cholestasis	1 day	<table><tr><td></td><td>1st</td><td>2nd</td><td>3rd</td></tr><tr><td>Urea</td><td>2.8-4.2</td><td>2.5-4.1</td><td>2.4-3.8</td></tr><tr><td>Creatinine</td><td>52-68</td><td>44-64</td><td>55-73</td></tr><tr><td>potassium</td><td colspan="3">3.3-4.1</td></tr><tr><td>Sodium</td><td colspan="3">130-140</td></tr><tr><td>Urate</td><td>0.14 - 0.23</td><td>0.14 - 0.29</td><td>0.21 - 0.38</td></tr></table>Normal ranges by trimester	Refer to obstetrician
Bilirubin/ bile acids (gold top)	Obstetric Cholestasis	1 day	<table><tr><td>1st</td><td>2nd</td><td>3rd</td></tr><tr><td>4-16</td><td>3-13</td><td>3-14</td></tr></table>Normal ranges by trimester	Refer to obstetrician
LFTs (gold top)	PET/ HELLP syndrome	1 day	<table><tr><td></td><td>1st</td><td>2nd</td><td>3rd</td></tr><tr><td>Total protein</td><td colspan="3">48-64</td></tr><tr><td>Albumin</td><td colspan="3">28-37</td></tr><tr><td>AST</td><td>10-28</td><td>11-29</td><td>11-30</td></tr><tr><td>ALT</td><td colspan="3">6-32</td></tr><tr><td>GGT</td><td>5-37</td><td>5-43</td><td>3-41</td></tr><tr><td>Alkaline phosphatase</td><td>32-100</td><td>43-135</td><td>133-418</td></tr><tr><td>Bile acids</td><td colspan="3">0-14</td></tr></table>Normal ranges by trimester	Refer to obstetrician
Urine PCR (White top urine)	Proteinuria	1 day	>0.3g	Refer to obstetrician
Clotting Platelets less than 100 (blue top)	DIC	1 day	Fibrinogen 3.63-4.23 Prolonged clotting times for prothrombin PT and activated oartial thromboplastin APTT	Refer to obstetrician
Thyroid Function Tests (gold top)	Thyroid	1 day	TSH normal range 0.03-4.0mIUL Repeat TFTs every 4 weeks in 1st trimester and at 16 and 28 weeks	Refer to obstetrician if results are outside this range
CRP (gold top)	context abdominal pain, infection PROM	1 day	0-7	Refer to obstetrician

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Appendix 2

Referral to Antenatal Screening Specialist for Positive Results or for Women who Decline Blood Test

Patient Details (use sticker if available)
Name: Contact number:
DOB:
Address:
Hospital No.

Date Booked:
Booked by (name of Midwife):
Team (tick as appropriate): Amber
Coral
Emerald
Sapphire
Hospital (OOA)

Date Haemoglobinopathy blood test taken: ----------
Or patient already aware

Date (tick as appropriate) HIV Hepatitis B Syphilis
Or patient aware of status
Date antenatal screening test taken

Date referred to Specialist Midwife:
Patient is a (tick as appropriate): Thalassaemia carrier
Sickle cell carrier (AS)
Hb C carrier (AC)
Hb D carrier (AD)
Hb E carrier (AE)
Raised Hb F (above 5%)
Other variant (please state)
Partner (tick as appropriate): never tested
Aware of status, tested on…….
Result: normal / unknown / carrier (state type)
Please place form in my tray in Antenatal Clinic
Helen Hosking (Specialist Midwife, Haemoglobinopathies)
Hillingdon Hospital direct and ask for ext. 2103 email: Helen.hosking@nhs.net

Date referred to Specialist Midwife for infectious diseases:
(Tick as appropriate) **HIV** **Hepatitis B** **Syphilis**
Direct number 01895 279768 email: Susie.phillips@nhs.net

Date referred to antenatal screening midwife:

Date referred for High Risk Scan:

Appendix 3
Hillingdon Antenatal Infectious Diseases Screening Pathway

1. Purpose and scope of the procedure

There is a requirement by the Antenatal clinic for a serology service for Infectious Diseases in Pregnancy Screening (IDPS). The guidelines state that samples will be processed urgently if a woman arrives un-booked in labour or late 3rd trimester. The results for HIV 2-4 hours and Hepatitis B and syphilis results would be done and available within 24 hours.

This document gives the procedure which should be followed for requesting urgent screening during and out of routine hours.

2. Personnel

It is the responsibility of all staff requesting screening tests to follow this procedure.

3. Related Documents

N/A

4. Procedure

- Take patient blood, requesting the appropriate urgent request codes on Cerner.
- Contact the Hillingdon Pathology team to arrange urgent testing:

N.B. the urgent test codes must only be used for urgent patients. Routine Antenatal codes must be used for routine patients.

Contact Details:

- **Normal Working Hours 09.00 – 17.00**
  - Pathology Laboratory (Hillingdon Biochemistry) **ext. 3228**
- **Out of hours 17.00 – 09.00 or at weekends:**
  - On Call Biochemist **Bleep 5602**

Send samples to the Hillingdon Blood Sciences Laboratory immediately for urgent testing.

- Hillingdon Laboratory team will arrange urgent processing and liaise with Infection & Immunity at Charing Cross and send samples for any confirmation testing required.
- Results available on Cerner

Follow up confirmation tests on screening test positive samples will be sent for additional testing within normal working hours.

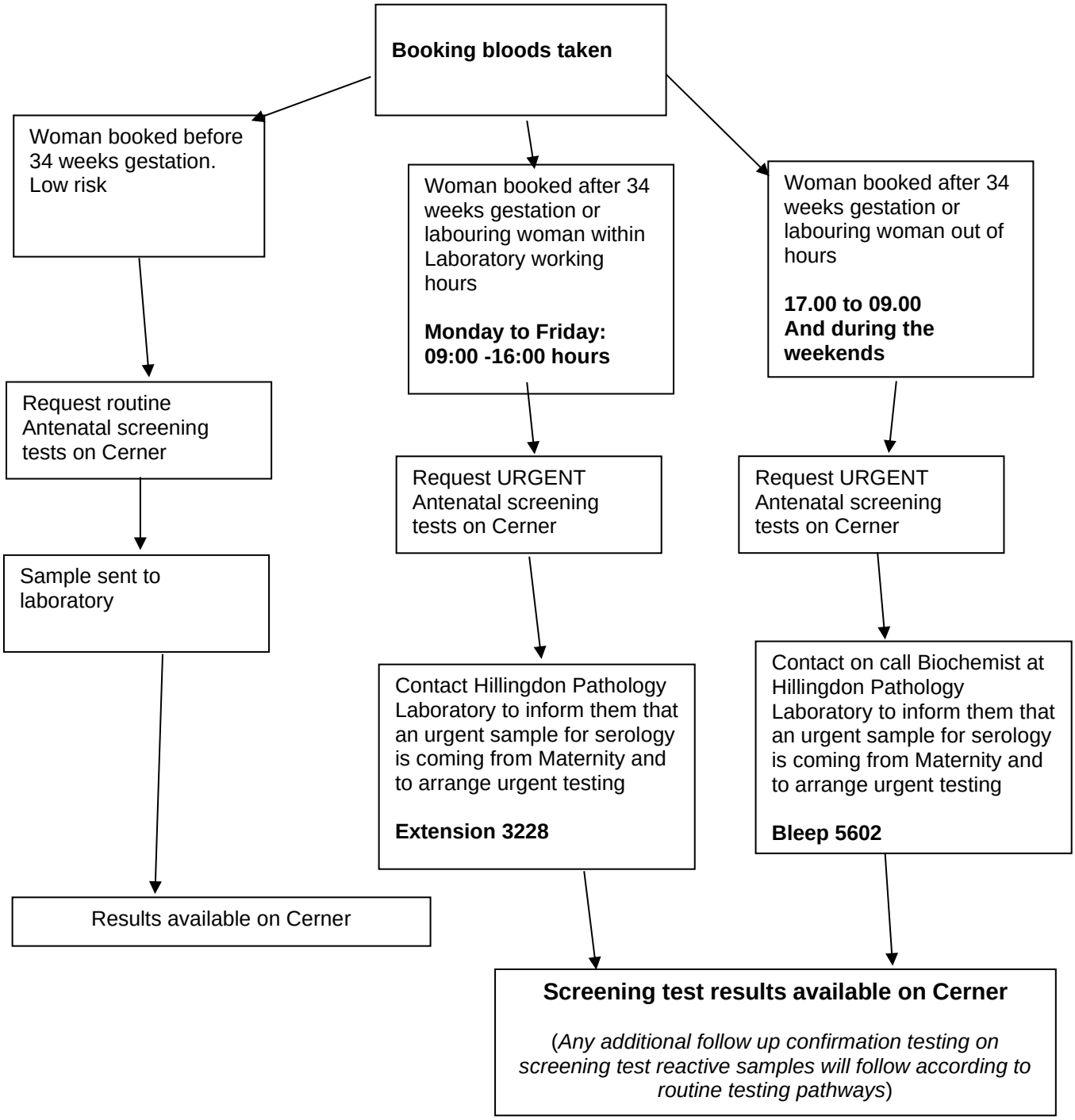
*Please note: it is critical that the above team is contacted by phone and the correct urgent Antenatal tests requests are made on Cerner in order to implement urgent testing pathways.

5. Patients Declining Tests

HIV

If patient declines to have HIV, Hepatitis B or Syphilis testing the Antenatal test concerned must not be ordered on Cerner. If any tests are ordered in error then the lab must be contacted immediately to inform them so it can be cancelled on the LIMS.

Appendix 4: Flow chart for Maternity bloods:



Appendix 5: Low PAPP-A results letter

LOW PAPP-A RESULTS Date:

As part of screening tests, levels of a protein developed in the placenta called PAPP-A (pregnancy-associated plasma protein A) are measured. Your level of PAPP-A is lower than average.

Some studies have found a small association between lower PAPP-A levels in early pregnancy and babies being smaller in the later stages of pregnancy. Only 1-2 pregnant women in every 100 with lower PAPP-A levels will be affected.

Although the chance of having a small baby when early PAPP-A levels are low is rare, the Royal College of Obstetricians and Gynaecologists recommend taking aspirin and closer monitoring during pregnancy.

We recommend that you take Aspirin 150mg once a day until 36 weeks of pregnancy. This helps increase the blood flow through the placenta and may reduce the chance of your baby being smaller. Please take this letter to any pharmacy and purchase aspirin. In rare circumstances, like an allergy, or liver or kidney disease, your GP may recommend that you do not take aspirin – if uncertain, please consult your GP.

We also recommend extra scans to monitor your baby's growth around 28, 32 and 36 weeks of pregnancy. Please take this letter to your next appointment with a midwife or obstetrician. They will discuss further and ask for the extra scans to be booked.

Receiving the news that you have a lower than average PAPP-A level can cause anxiety.

Please be reassured that most babies will have normal growth and you will be monitored closely by your midwife and obstetrician.

If you have any questions, please do not hesitate to contact your midwife or obstetrician.

Yours sincerely,

Midwife/Doctor/Consultant
Antenatal Clinic